Relations and orders solutions

DM07-E01

Loops rule out irreflexivity. Opposite pairs 1R2,2R1 with distinct endpoints rule out antisymmetry. R is reflexive, symmetric, and transitive: both blocks {1,2},{3} contain every internal pair. It is already an equivalence relation, so its smallest equivalence superrelation is itself, with those two classes.

DM07-E02

With indices a,b,c, source rows and destination columns give

$$M = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

The first row's ones are a→a,a→b. For example,

$$(M^{\odot 2})_{ac} = M_{aa}M_{ac} \vee M_{ab}M_{bc} = 0 \vee 1 = 1,$$

requiring a→c. The complete Boolean square is

$$M^{\odot 2} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix}.$$

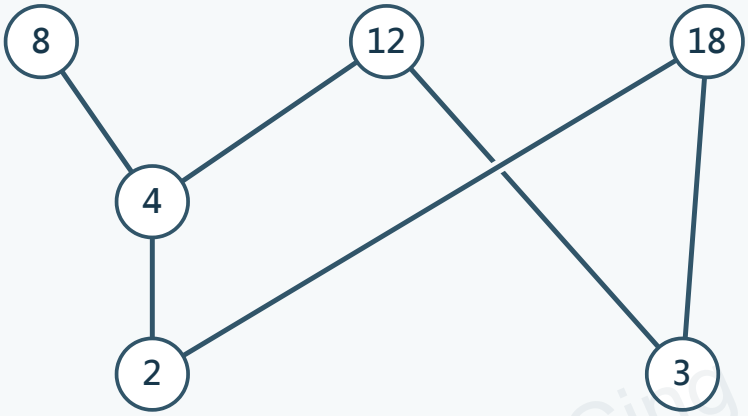
New pairs are (a,c),(b,a),(c,b). The pairs (b,b),(c,c) are not two-step reachable; they first arise from b→c→a→b and c→a→b→c. Thus $R \cup R^2 \cup R^3 = A^2$, the complete relation on three points.

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DM07-E03

Minimal elements are 2,3; maximal elements are 8,12,18. There is no minimum because 2,3 are incomparable, and no maximum because the three distinct maximal elements are incomparable.

Covers are 2→4,2→18,3→12,3→18,4→8,4→12. Neither 2→8 nor 2→12 is a cover because 4 intervenes. Possible intermediate divisors 6,9 for 2→18 lie outside S, so this remains a cover in the induced subposet.



An upward path means divisibility; only covers are drawn.

Cover edges
2–4 2–18 3–12
3–18 4–8 4–12

Minimal: 2, 3
Maximal: 8, 12, 18
No minimum or maximum

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DM07-E04

Every element belongs to a block, hence relates to itself. Exchanging a same-block pair proves symmetry. If a,b share B and b,c share C, the blocks intersect at b. A partition forces B=C, so a,c share a block, proving transitivity. The class of a consists of exactly all members of its block, recovering the original blocks.

DM07-E05

No. Take alphabet $\{0, 1\}$ and states p, q, t, only t accepting, with initial partition $\{p, q\}, \{t\}$:

State	0	1	Accepting?
p	p	t	No
q	p	q	No
t	t	t	Yes

For input 0, p and q reach the same block. Input 1 reaches accepting t from p and rejecting q from q, so the word 1 distinguishes them and they cannot merge.

A partition is stable if each block has a common acceptance mark and every two states in it transition into the same block for every input. Induction on suffix length then proves that states in one block have equal future acceptance, so merging them is safe. This is a sufficient global test, not a necessary condition for the current partition to establish that a particular pair is equivalent. To compare two DFAs, use the same alphabet and disjoint copies of their states before refining together.

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DM07-E06

Yes. Each diagonal pair belongs to both orders, hence the intersection. Opposite pairs in the intersection belong to each order, whose antisymmetry forces equality. Two composable pairs in the intersection lie in both orders; each order's transitivity gives the composed pair, which therefore belongs to their intersection. Totality is unnecessary. Removing comparisons this way preserves every inference whose premises remain.

DM07-E07

Reflexivity fixes n diagonal pairs and leaves n^2-n free, giving $2^{n(n-1)}$. Symmetry gives independent diagonal choices and one joint choice per unordered distinct pair, giving $2^{n+\binom{n}{2}}$. Antisymmetry permits neither direction, left, or right per distinct pair and free diagonal, giving $2^n 3^{\binom{n}{2}}$. Asymmetry forbids diagonals too, leaving $3^{\binom{n}{2}}$.

At n=3 the answers are 64,64,216,27. A symmetric antisymmetric relation is any diagonal subset, giving $2^3 = 8$, not just the empty relation. Reflexivity would leave equality alone.

DM07-E08

Initial rows are 0100/0010/0001/0000. Intermediate a adds nothing. Intermediate b adds (a,c), giving 0110/0010/0001/0000. Intermediate c adds (a,d),(b,d), giving 0111/0011/0001/0000. Intermediate d adds nothing. These six strictly upper-triangular pairs form R^+; add four diagonal pairs for R^*.

For cycle a→b→c→a, R^2 contains only (a,c),(b,a),(c,b). Union with R still lacks all diagonal pairs. R^3 supplies (a,a),(b,b),(c,c), proving positive reachability may require n steps.

DM07-E09

Ignoring direction gives components {1,2,3},{4}. The closure is $\{1,2,3\}^2 \cup \{(4,4)\}$, with ten pairs. The same-block definition proves equivalence. Any equivalence superrelation must contain 2R1,2R3 by symmetry, then 1R3,3R1 by transitivity, and every diagonal by reflexivity. It therefore contains all ten pairs, proving minimality.

Original R is already transitive: 2 has no outgoing edge, so no composable two-edge path exists. Adding inverses and diagonals alone still lacks (1,3),(3,1); another transitive closure is necessary.

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DM07-E10

For S={2,3}, lower bounds are {1}, upper bounds {6,12}, glb 1, lub 6. For T={4,6}, lower bounds are {1,2}, upper bounds {12}, glb 2, lub 12. Search for bounds in the full six-element poset, not just inside S or T.

Prefix word	Linear extension
AAABBB	1,2,4,3,6,12
AABABB	1,2,3,4,6,12
AABBAB	1,2,3,6,4,12
ABAABB	1,3,2,4,6,12
ABABAB	1,3,2,6,4,12

A is chain 1,2,4; B is chain 3,6,12. Their internal orders are fixed, and the jth B requires the jth A earlier. Thus extensions biject with three-pair Catalan words, and these five exhaust them.

Deleting 1,4 leaves {2,3,6,12}. Elements 2,3 have no common lower bound, so their glb disappears while lub 6 remains. Since T contains deleted 4, it is no longer a subset and its bounds are no longer a valid question in this poset.

DM07-E11

From any x in a finite nonempty poset, strictly ascend until maximal. Vertices cannot repeat, so termination is forced. Uniqueness of maximal m makes every ascent end at m, proving $x \preccurlyeq m$ for every x. For an infinite counterexample, add isolated z to the natural-number chain. Only z is maximal, but it is not above any natural number.

A finite nonlattice is {0,a,b,u,v,1}, with covers 0–a,0–b,a–u,a–v,b–u,b–v,u–1,v–1. Bottom is 0 and top 1. Common upper bounds of a,b are u,v,1. The incomparable u,v are both minimal upper bounds but neither is least, so $a \vee b$ does not exist. An arbitrary minimal upper bound cannot replace the lub.

DM07-E12

Every integer partition of five gives the following squared-size sum.

Class sizes	Pair count
5	25
4+1	17
3+2	13
3+1+1	11
2+2+1	9
2+1+1+1	7
1+1+1+1+1	5

Each size pattern is realizable, so the possible counts are exactly {5,7,9,11,13,17,25}. The complete list excludes other odd counts such as 15,19,21,23.

Choose a nonempty proper subset B and use its complement as the other block, in $2^5 - 2$ ways. B and its complement describe the same partition, are distinct, and count each partition exactly twice. Division by two gives 15, also S(5,2).

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DM07-E13

P_0={A,B,C,D}/{E,F}. The 0/1 successor-acceptance signatures are (no,no) for A,D and (no,yes) for B,C, giving P_1={A,D}/{B,C}/{E,F}.

Relative to P_1, A has successor labels (BC,BC), D (AD,AD), B,C both (AD,EF), E,F both (EF,EF). Thus P_2={A}/{D}/{B,C}/{E,F}. Relative to P_2, B,C both have (D,EF) and E,F both (EF,EF), so no further split occurs and P_3=P_2.

Write U={A},V={B,C},D={D},T={E,F}.

State	0	1	Accepting?
U	V	V	No
V	D	T	No
D	D	D	No
T	T	T	Yes

All six unordered class pairs have witnesses: UT,VT,DT use ε; UV,VD use 1; UD uses 01. All four classes are reachable and every pair has different future acceptance. A same-language DFA with fewer than four states must send two of their four access prefixes to one state. Appending their witness would then give equal acceptance there but opposite acceptance here, contradiction. Four states are necessary.

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DM07-E14

P_0={p,q,r,s}. Output rows p:(0,0), q/r:(0,1), s:(1,1) give P_1={p}/{q,r}/{s}. Successor signatures (qr,qr),(qr,s),(s,s) remain constant within blocks, so $P_2=P_1$.

Quotient state	Input 0 next/output	Input 1 next/output
{p}	{q,r}/0	{q,r}/0
{q,r}	{q,r}/0	{s}/1
{s}	{s}/1	{s}/1

Input 010 from p follows p --0/0→q --1/1→s --0/1→s and emits 011. Added t matches p on every one-symbol output and initially shares its P_1 block. On zero, p reaches q with output row (0,1), t reaches s with row (1,1); the next round separates them. Witness 00 gives outputs 00,01. Length zero and one always agree, establishing shortestness. If t is unreachable from initial p, remove it before initial-state minimization; if t is another possible initial state, keep its distinct class.

DM07-E15

Integer-valued a^2 is not well-defined: [1]=[6] but $1 \neq 36$. Class-valued $[a^2]$ is well-defined because $5 \mid (a-b)$ implies $5 \mid (a-b)(a+b) = a^2 - b^2$, hence $[a^2] = [b^2]$. The third is not well-defined: [0]=[5] but $(-1)^0 = 1$, $(-1)^5 = -1$. Failure concerns different representative values for one input class, not differing codomain notation.

References and attribution

Lin Cing, handwritten discrete mathematics notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Ch. 7.

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